TRACE.ai

Transaction Risk Analysis and Compliance Engine

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System Overview

TRACE.ai is structured as five independent layers, each exposing a clean interface to the next. No layer imports the internals of another, allowing any module to be replaced independently. The system ingests synthetic banking transaction data (IBM AMLSim), models it as a directed graph, runs four fraud detection signals in parallel, fuses their scores, and serves results to a React 3D dashboard over FastAPI and WebSocket.

Architecture Diagram

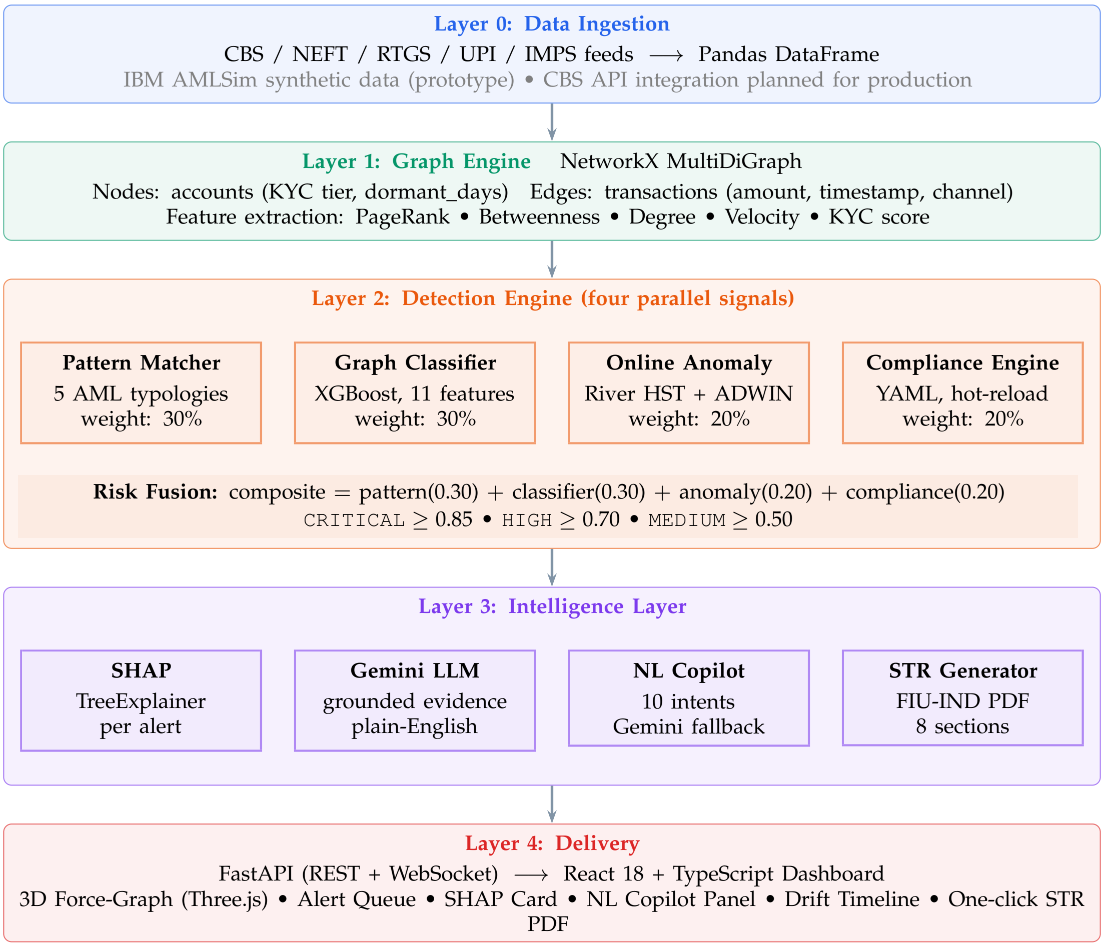

Figure 1: TRACE.ai five-layer architecture with one-directional data flow. Each layer exposes a clean interface; no layer imports the internals of another.

Technology Stack

Layer	Technology	Why This Choice
Data	IBM AMLSim + Pandas 2.0	Realistic synthetic Indian banking data; reproducible generation; no real CBS required.
Graph Engine	NetworkX 3.3 MultiDiGraph	Sufficient for 500 to 5,000 account prototype; no external database. Neo4j GDS targeted for production.
Graph Features	scikit-learn + NetworkX	11 features (degree, PageRank, betweenness, velocity, KYC score, dormancy) capturing graph structure that a GNN would learn from raw adjacency.
ML Classifier	XGBoost 2.1.4	AUC above 0.99 on synthetic data; natively supported by SHAP TreeExplainer; interpretable for compliance teams.
Online Anomaly	River 0.23.0 (HST + ADWIN)	Per-account streaming baselines at sub-millisecond latency; no batch retraining needed.
Compliance	PyYAML + watchdog	Hot-reload YAML rules without restart; compliance teams update rules without engineering involvement.
LLM	Gemini 2.5 Flash (Vertex AI)	GCP credits available; grounded evidence block prevents hallucination; fast for structured prompts.
STR PDF	ReportLab 4.1	Programmatic 8-section A4 PDF in FIU-IND format with tables, fund trail, and risk charts.
Backend	FastAPI 0.110 + WebSocket	Asynchronous; WebSocket for live alert streaming; OpenAPI docs auto-generated.
Frontend	React 18 + TypeScript + Vite	Type-safe hooks architecture; Vite proxy for local dev; compiled SPA served by FastAPI.
3D Graph	react-force-graph-3d + Three.js	GPU-accelerated force-directed graph; nodes glow by risk level in real time.
Deploy	GCP Cloud Run + Docker	Single container; no CORS configuration; scales to zero when idle; demo data baked in at build time.

Key Technical Decisions

- **XGBoost instead of a full Temporal GNN.** Phase 1 described ChronoWave-GNN. For the prototype, XGBoost on graph-structural features is the correct approach: PyTorch Geometric TGN requires significantly more data and compute. SHAP TreeExplainer provides attribution equivalent to GNNExplainer. A trained TGN checkpoint (`models/tgn_classifier.pt`, AUC 0.72) is included; trade-offs documented in `MODEL_CARD.md`.
- **NetworkX instead of Neo4j.** Adequate for 500 to 5,000 accounts; eliminates an external dependency. The graph builder (`src/trace/graph/builder.py`) can be swapped for a Neo4j connector without touching detection or intelligence layers (ADR-0003 in `DECISIONS.md`).
- **Compliance rules as YAML, not code.** RBI Master Directions change frequently. YAML hot-reload with watchdog lets compliance teams update rules without an engineering deployment. A new circular takes effect within 5 seconds.
- **Single-container deployment.** FastAPI serves the React SPA as StaticFiles. One Cloud Run URL; no CORS configuration. Demo data is seeded at image build time so the container starts with all alerts pre-populated.

Known Limitations

- **Synthetic data only.** Metrics are on IBM AMLSim data. Real-bank performance requires retraining on CBS feeds.
- **Batch graph, not a real-time stream.** Production requires a Kafka consumer for live CBS event ingestion.
- **NetworkX at prototype scale.** Not suitable for millions of accounts; Neo4j GDS is the production path.
- **No user authentication.** Dashboard is publicly accessible (POC). Production requires LDAP/SSO.
- **FIU-IND STR format is simulated.** The PDF follows the 8-section structure but is not legally compliant and has no digital signature.
- **5 fraud typologies in POC.** Production requires 15 to 20; adding a new typology requires only a new module in `src/trace/detection/patterns/`.

Live: `https://trace-ai-4xnj5ovp4a-uc.a.run.app` **Repo:** `https://github.com/wildcraft958/namofans-trace-ai`